

Laboratory Requisition — Ballista Investigation

Unit: Forces & Motion (Yr 7–8)

Requested

by: Date needed: _____

Lesson: Projectile investigation

Number of groups: _____

Period / room: _____

What the students are doing

Students fire a projectile from a ballista at a **measured launch angle** from ground level, then film each shot in slow motion. From the video they measure the total time of flight and the maximum height reached, and from the floor they measure the horizontal range (R). They use this to calculate gravity (g). Each group runs **10 shots**, so allow 50–60 minutes. *Note: unlike the trebuchet (high arc, horizontal launch), the ballista is fired at an adjustable angle from the ground.*

Equipment required

☐	Item	Per group	Total needed	Notes
☐	Ballista (torsion or elastic)	1	_____	Class set, or shared between groups
☐	Projectile	1	_____	40 g foam dart or soft ball; identical for all groups
☐	Spare projectiles	2	_____	Same mass — in case of loss
☐	Protractor (large) or angle gauge	1	_____	To set & read the launch angle
☐	Measuring tape (5 m+)	1	_____	For measuring range R
☐	Metre ruler	1	_____	For scale reference in video
☐	Spirit level	1	_____	To check the ballista base is level
☐	Masking tape	1 roll	_____	Mark launch line + exclusion zone
☐	Smartphone or tablet + tripod	1	_____	Must shoot slow motion (≥ 120 fps)
☐	Safety glasses	4	_____	One per student — mandatory

Setup before lesson

- Place each ballista on a level surface at one end of a clear run; check the base with a spirit level.
- Mark the **launch line** on the floor with tape, directly below the projectile’s starting position. Label it “**0 m**”.
- Lay the measuring tape along the floor from the “0 m” line toward the landing zone.
- Attach or stand a protractor at the pivot so students can set and read the launch angle.
- Place a metre ruler upright near the flight path (visible in video) as a scale reference.
- Mark a **2 m exclusion zone** in front of each ballista with tape.
- Confirm each camera is charged and slow-motion mode is on.

Safety

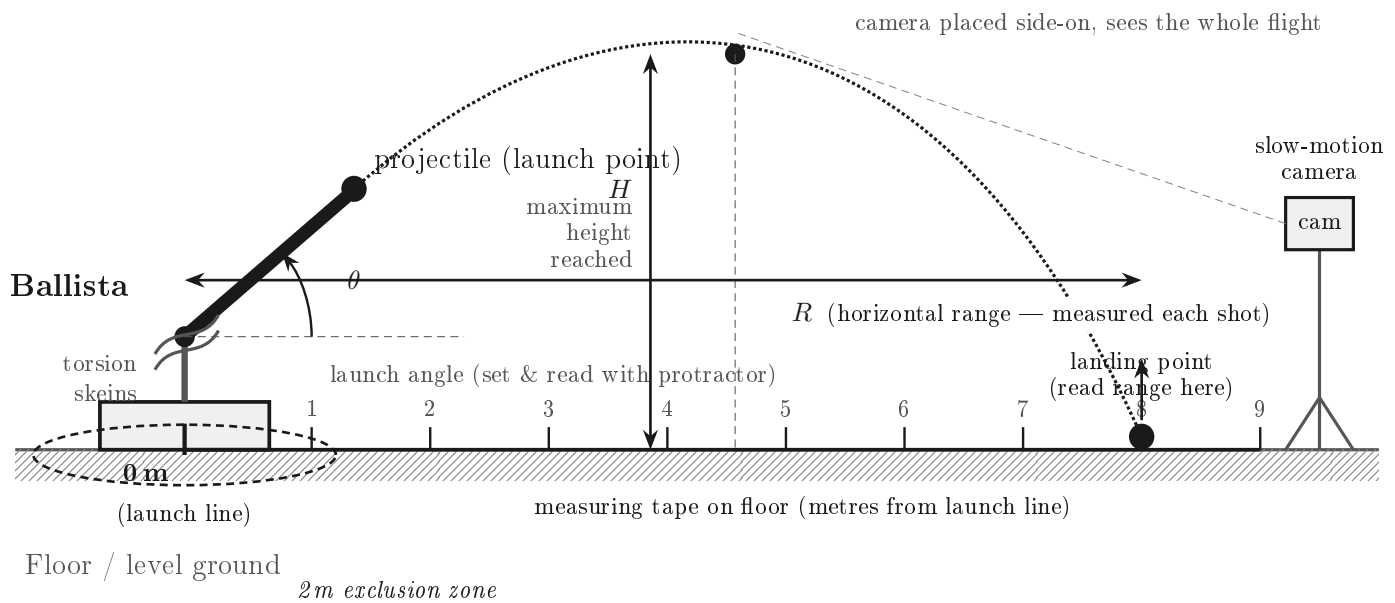
- **Risk level: Medium.** Teacher supervises all shots.
- Use only **soft / foam projectiles** — never hard or pointed objects.
- Safety glasses worn by all students during firing.
- Only one person fires each ballista; others stand **behind** it.
- Nobody in front of or beside the firing line.
- Never aim toward people — aim down the marked range only.
- Projectiles retrieved only on teacher’s “all clear”.
- Firing mechanism reset and locked before reloading.

Notes / queries for teacher: _____

Prepared by (tech): _____

Ballista — Station Setup Diagram

Plan view of one ballista station. Set up each station as shown below. All distances are examples — actual values depend on the ballista and launch angle used.



Key to the diagram

θ	Launch angle — set and read with a protractor mounted at the pivot. This is the variable students change between sets of shots.
H	Maximum height the projectile reaches — read from the slow-motion video against the upright metre ruler used as a scale reference.
R	Horizontal range — distance from the “0 m” launch line to where the projectile lands, read off the measuring tape after each shot.
Camera	Positioned side-on at a fixed distance so the whole flight path is in frame. Students count video frames to find the time of flight and the time to reach peak height.
Skeins	The twisted rope bundles (or elastic) that store the launching energy. Keep tension/twist identical between shots within a group.

One station shown. Replicate this layout for each group, leaving clear space between stations and down each firing lane.